MTH207

Timed Online Assignment – July Semester 2021

Linear Algebra

Tuesday, 16 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Let V=$\left\{\begin{bmatrix} a+2b-c \\ 2a-b+8c \\ -a+3b-9c \\ 3a+2b+5c \end{bmatrix} : a,b,c \in \mathbb{R}\right\}$ be a set in $\mathbb{R}^4$.

Let S= $\left\{\begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 8 \\ -9 \\ 5 \end{bmatrix}\right\}$ be a set of three vectors in $\mathbb{R}^4$.

(a) Determine whether S is a linearly independent set. If the set S is not linearly independent, find a linear relationship among them.

(8 marks)

(b) Show that V is a subspace of $\mathbb{R}^4$ and state the dimension of V.

(6 marks)

(c) Does $\begin{bmatrix} 2 \\ -6 \\ 8 \\ -2 \end{bmatrix}$ belong to V?

(3 marks)

(d) Explain whether $\left\{\begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 8 \\ -9 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ -6 \\ 8 \\ -2 \end{bmatrix}\right\}$ forms a basis of $\mathbb{R}^4$.

(3 marks)

Question 2

(a) Compute the row echelon form of the following matrix. Assume $a \neq 1$.

$$\begin{bmatrix} 4 & 2 & 1 & 3 & 0 \\ 6 & 3 & 4 & 7 & 0 \\ 2 & 1 & 0 & a & 0 \end{bmatrix}$$

(10 marks)

(b) Hence, or otherwise, find the solution set for the following equations:

$$\begin{aligned} 4x + 2y + z + 3w &= 0 \\ 6x + 3y + 4z + 7w &= 0 \\ 2x + y \qquad\quad + \ w &= 0. \end{aligned}$$

(10 marks)

Question 3

(a) Calculate all possible values of k for which A= $\begin{bmatrix} 1 & 3 & k \\ -1 & -1 & 1 \\ 4 & 7 & 5 \end{bmatrix}$ is invertible.

(8 marks)

(b) If the matrix in Question 3(a) is invertible and

$$\begin{bmatrix} 1 & 3 & k \\ -1 & -1 & 1 \\ 4 & 7 & 5 \end{bmatrix}^{-1} = \frac{1}{(5-k)} \begin{bmatrix} -4 & \frac{1}{3}(7k-15) & \frac{1}{3}(k+3) \\ 3 & \frac{1}{3}(5-4k) & \frac{-1}{3}(k+1) \\ -1 & \frac{5}{3} & x \end{bmatrix}.$$

Find the value of x.

(6 marks)

(c) Hence, implement matrix algebra to solve the following system of linear equations:

$$\begin{aligned} x + 3y + 4z &= 1 \\ x + y - z &= -3 \\ 4x + 7y + 5z &= 3. \end{aligned}$$

(6 marks)

Question 4

Define T: $\mathbb{R}^3 \to \mathbb{R}^3$ by

$$T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x+3y+4z \\ 3x+4y+7z \\ -2x+2y \end{bmatrix}.$$

You may assume T is a linear transformation.

(a) Compute the matrix representation $\boldsymbol{A}$ of T with respect to standard basis in domain and codomain.

(3 marks)

(b) Compute a basis for *Col* ($\boldsymbol{A}$), the column space of $\boldsymbol{A}$ and the dimension of *Nul* ($\boldsymbol{A}$), the null-space of $\boldsymbol{A}$.

(10 marks)

(c) Is $\begin{bmatrix} 2 \\ 2 \\ -2 \end{bmatrix}$ in *Nul* ($\boldsymbol{A}$)? Is $\begin{bmatrix} 5 \\ 5 \\ 6 \end{bmatrix}$ in *Col* ($\boldsymbol{A}$)?

(3 marks)

(d) Give a geometrical description of *Nul* ($\boldsymbol{A}$) and *Col* ($\boldsymbol{A}$) in $\mathbb{R}^3$.

(4 marks)

Question 5

Let $\boldsymbol{A}$ be the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

Denote vectors $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ by $\boldsymbol{e}_1$ and $\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ by $\boldsymbol{e}_2$.

(a) Show that $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$ are eigenvectors of A and calculate the eigenvalues corresponding to $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$.

(6 marks)

(b) Determine a remaining eigenvector $\boldsymbol{e}_3$ such that $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ form a linearly independent set of eigenvectors of $\boldsymbol{A}$.

(6 marks)

(c) Solve for an invertible matrix P and a diagonal matrix D such that

$$\mathrm{P}^{-1} A \, \mathrm{P} = D \ .$$

(3 marks)

(d) Is $\mathrm{P}\, A^{-1}\, \mathrm{P}^{-1}$ well-defined? Explain.

(5 marks)

----- END OF PAPER -----